

Course: Differential Geometry

Assignment Number 2

1. Prove that given any n dimensional manifold M , the tangent bundle TM is always orientable.(2 marks)
2. Find all the geodesics on the cylinder $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1\}$.(3 marks)
3. Prove the existence and uniqueness of the Riemannian connection on any Riemannian manifold.(3 marks)
4. Let P, Q and R be smooth functions in $\mathbb{R}^3$ such that for all $p \in \mathbb{R}^3$ atleast one of $P(p), Q(p)$ or $R(p) \neq 0$. Define the distribution $\mathcal{D} \equiv (P \frac{\partial}{\partial x} + Q \frac{\partial}{\partial y} + R \frac{\partial}{\partial z})^\perp$ for all $p \in \mathbb{R}^3$. Derive a sufficient condition on the functions P, Q and R such that the distribution $\mathcal{D}$ is integrable.(2 marks)